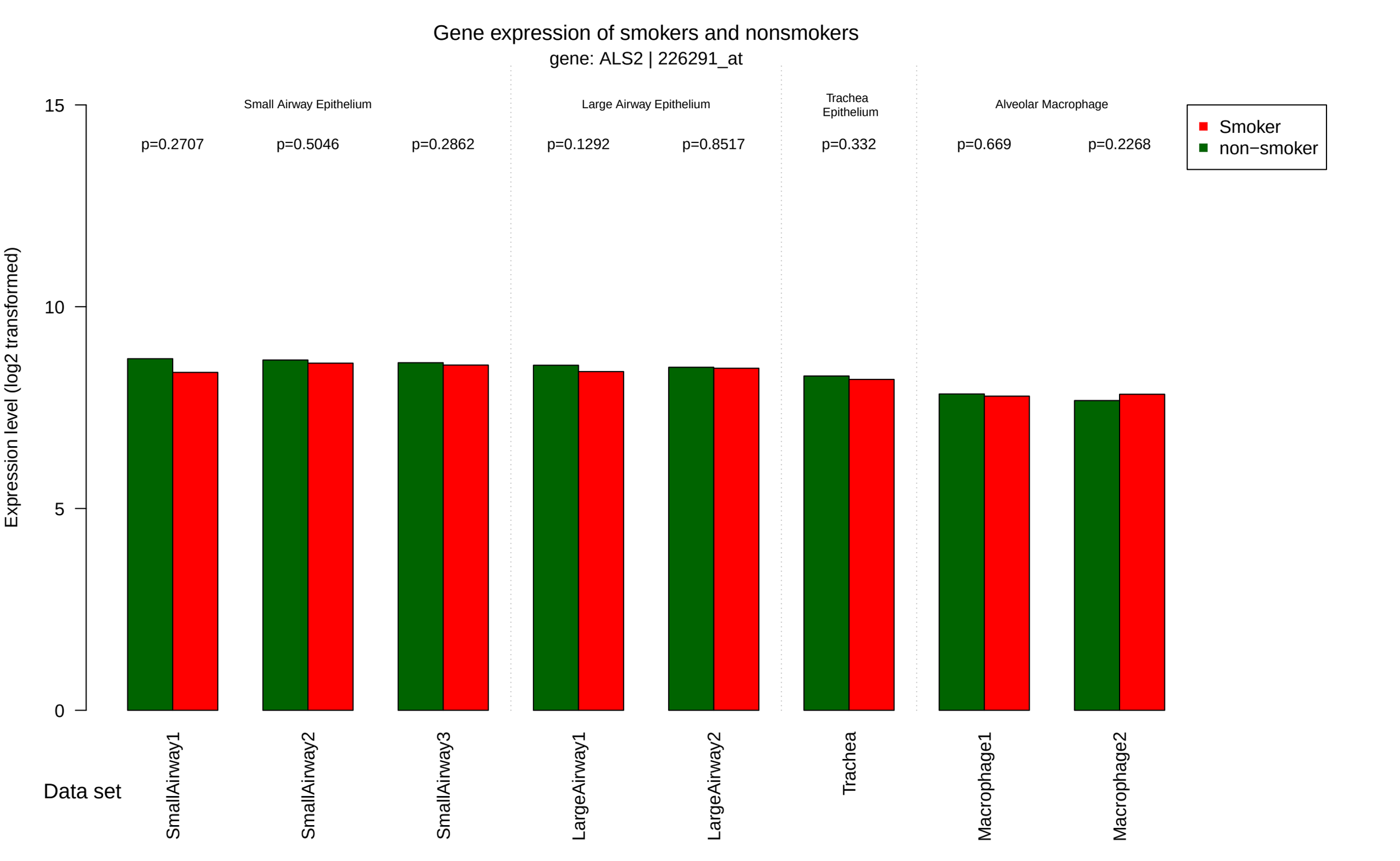
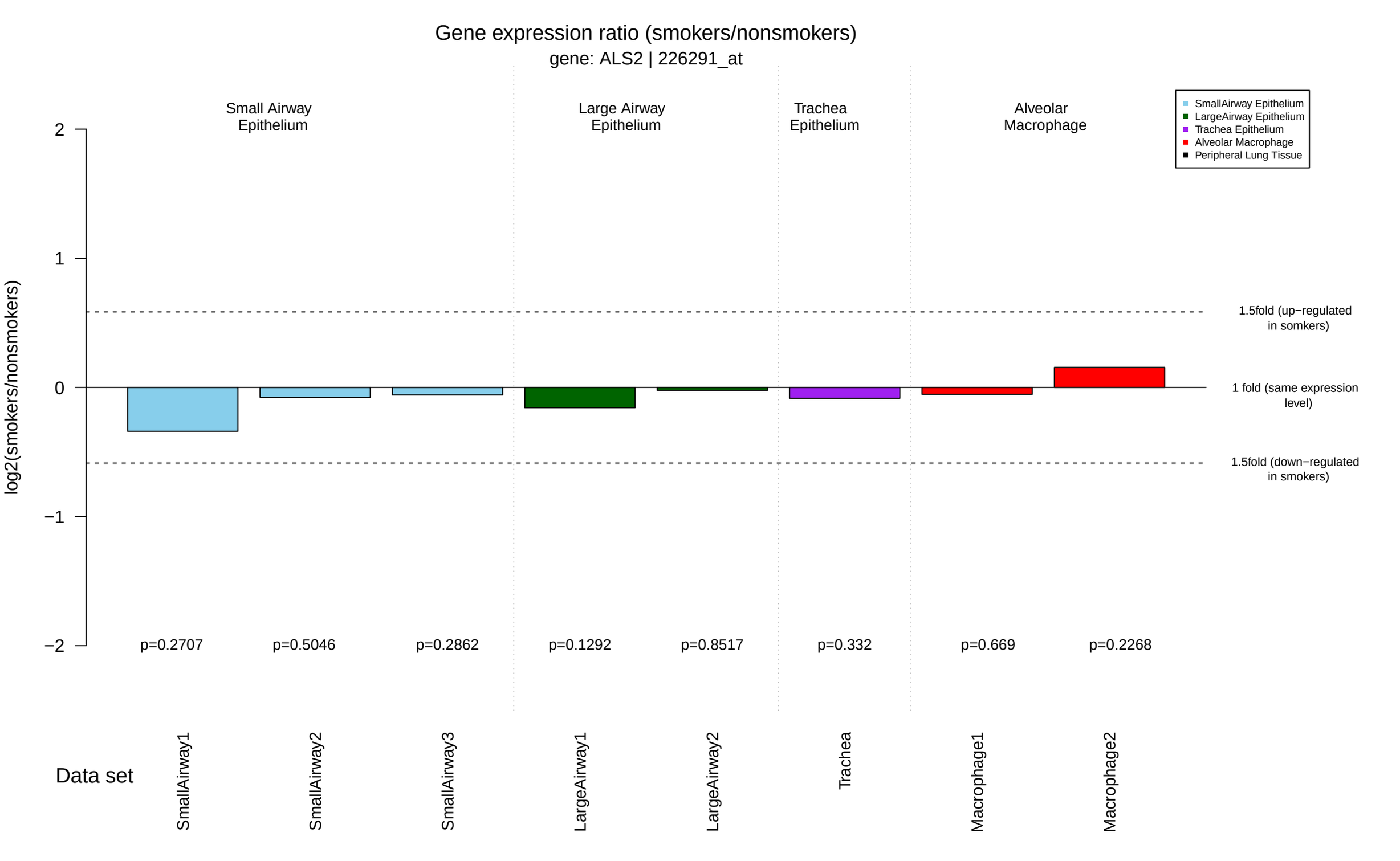
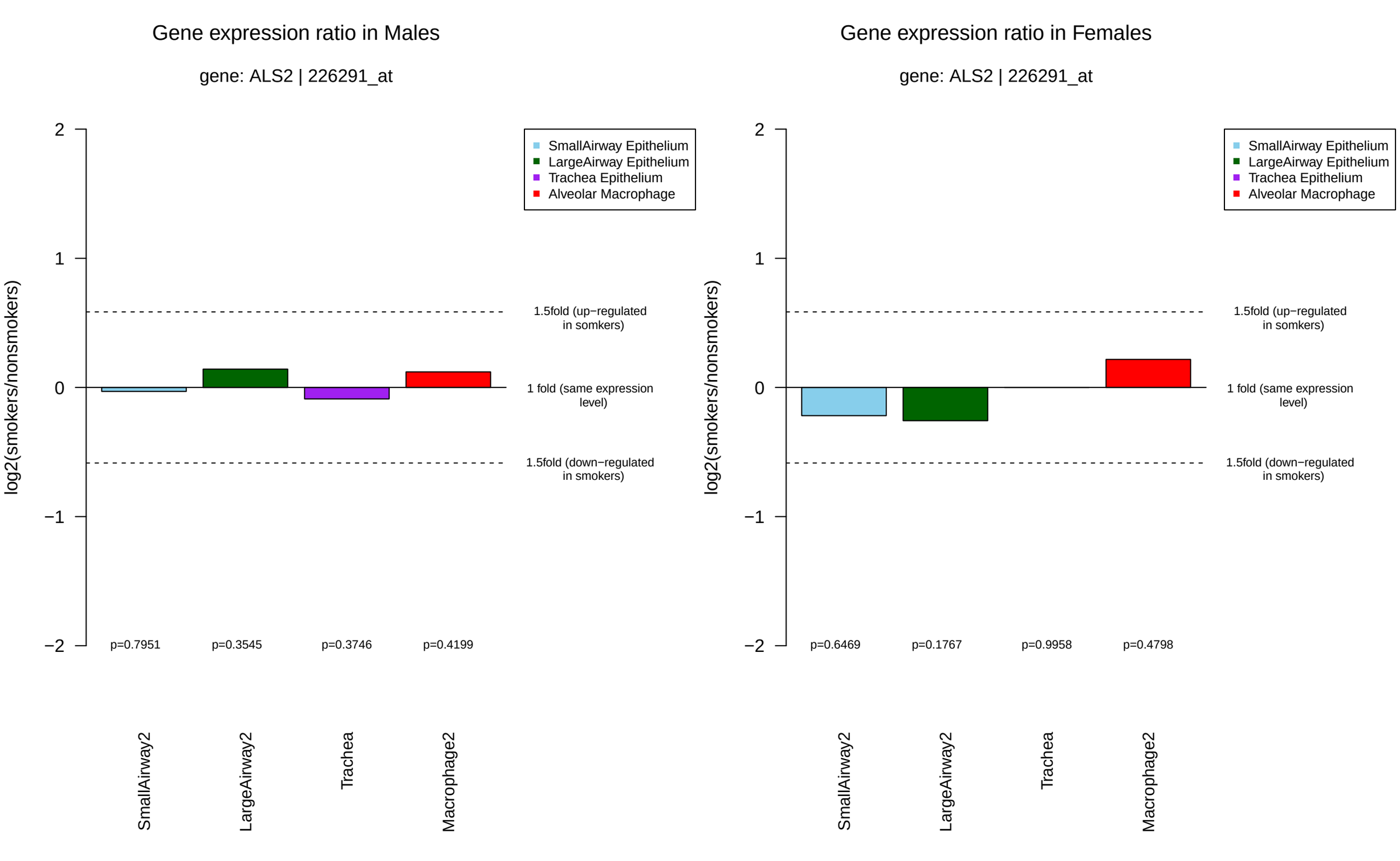
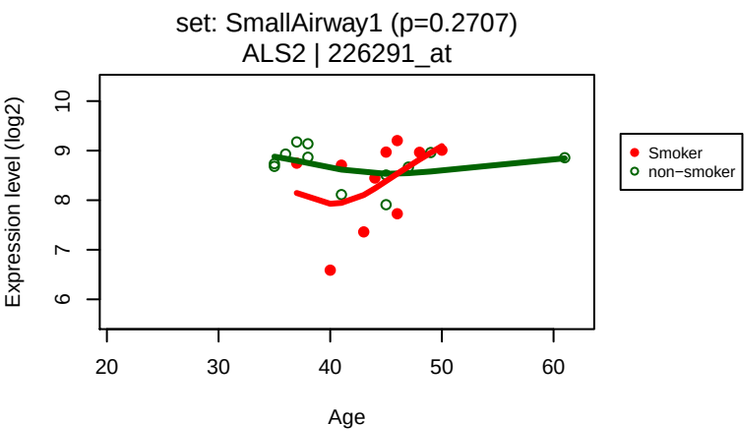
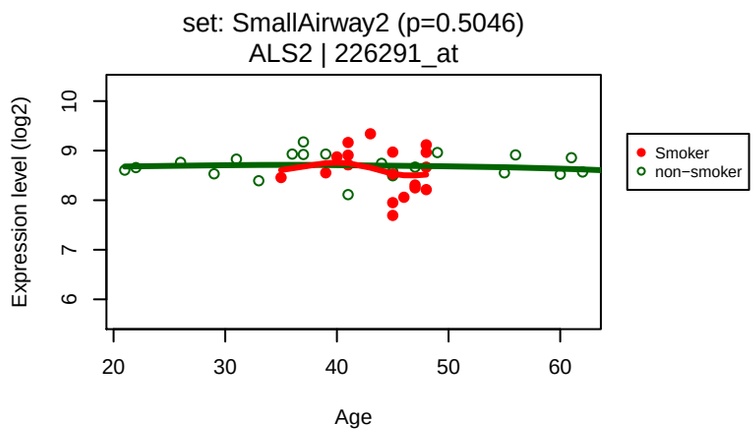
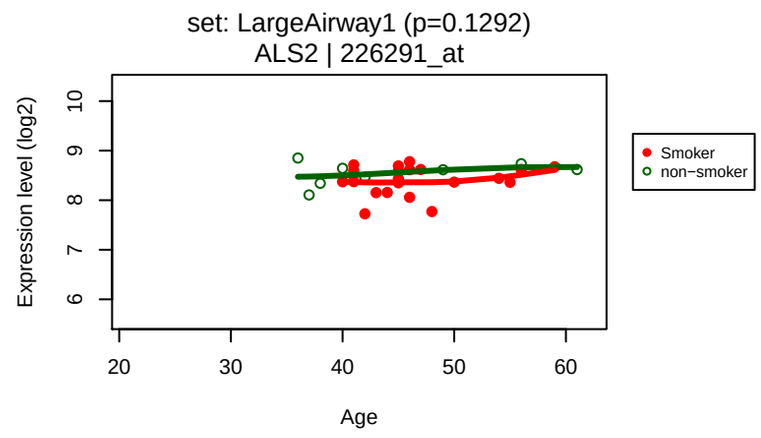

Gene expression of smokers and nonsmokers
gene: ALS2 | 226291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2707
p=0.5046
p=0.2862
p=0.1292
p=0.8517
p=0.332
p=0.669
p=0.2268
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

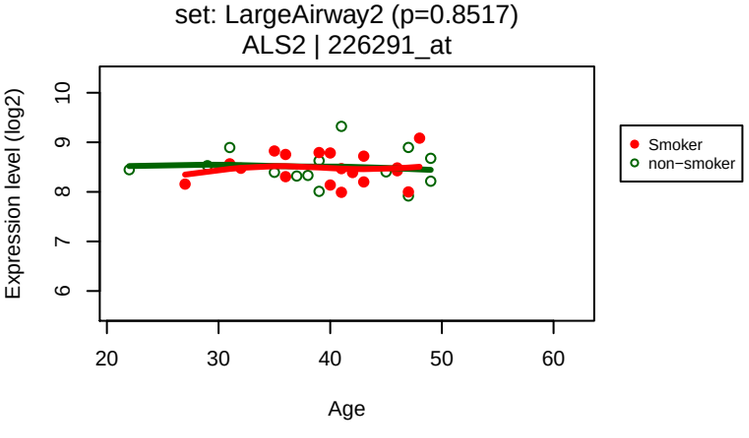

Gene expression ratio (smokers/nonsmokers)
gene: ALS2 | 226291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2707
p=0.5046
p=0.2862
p=0.1292
p=0.8517
p=0.332
p=0.669
p=0.2268
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

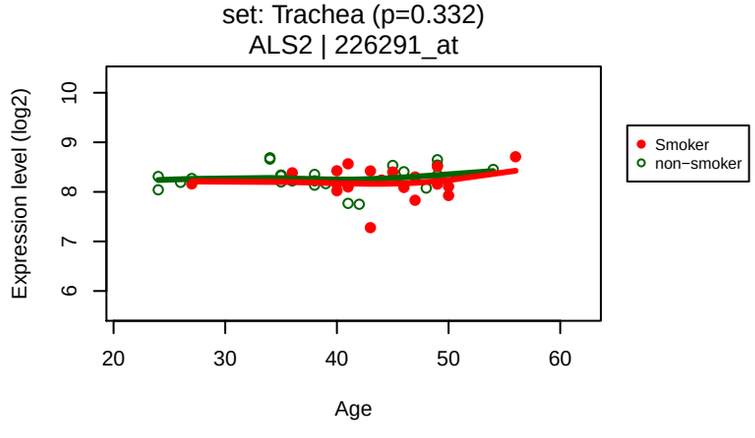

Gene expression ratio in Males
gene: ALS2 | 226291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7951
p=0.3545
p=0.3746
p=0.4199
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALS2 | 226291_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6469
p=0.1767
p=0.9958
p=0.4798
SmallAirway2
LargeAirway2
Trachea
Macrophage2

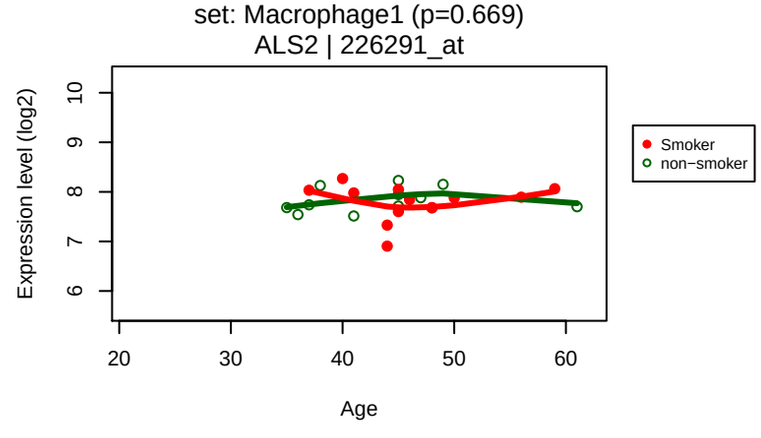

set: SmallAirway1 (p=0.2707)
ALS2 | 226291_at
Expression level (log2)
Age
Smoker
non−smoker

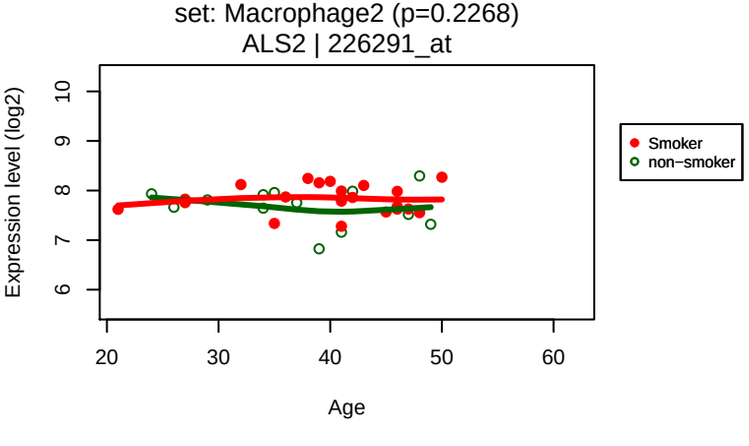

set: SmallAirway2 (p=0.5046)
ALS2 | 226291_at
Expression level (log2)
Age
Smoker
non−smoker

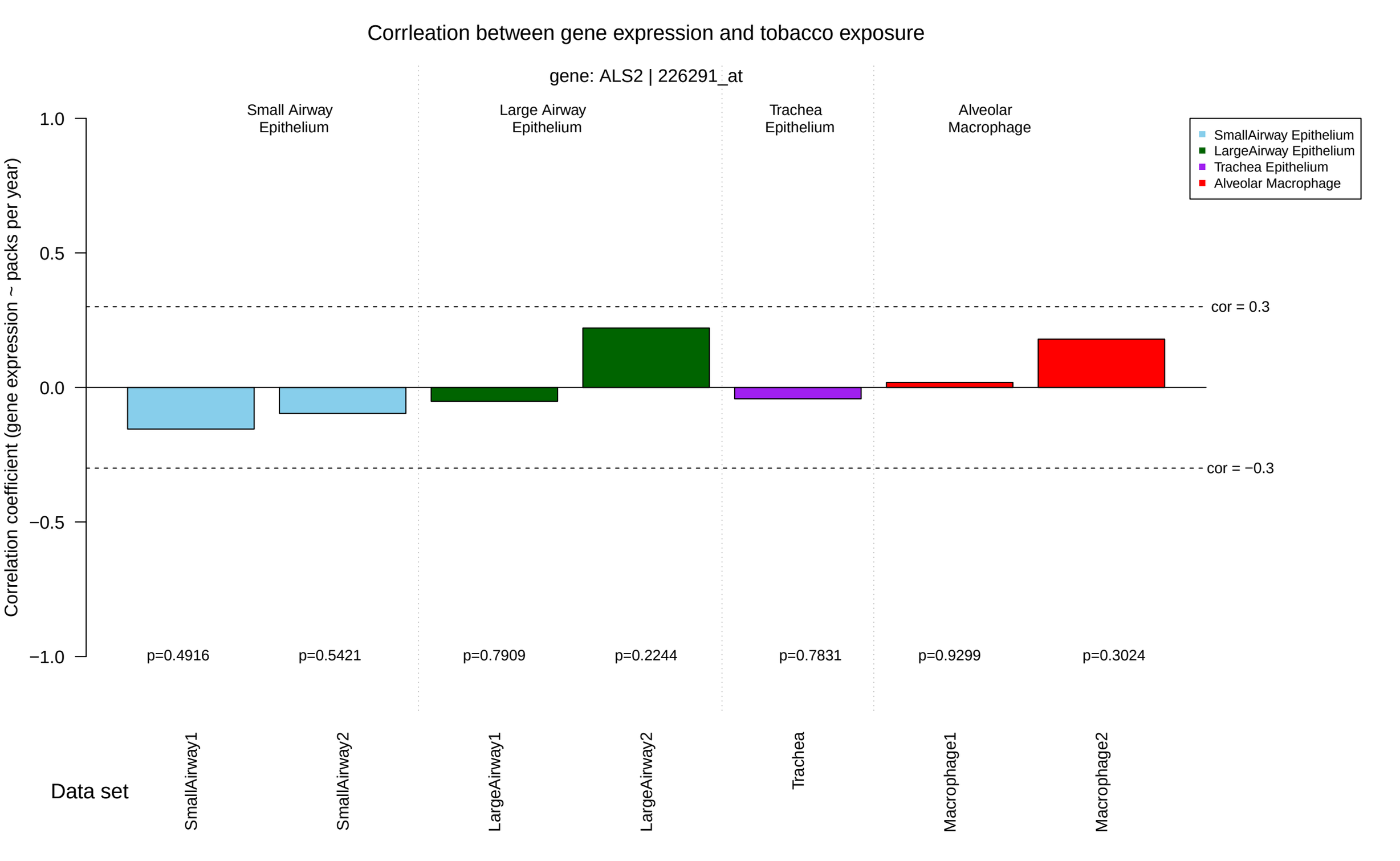

set: LargeAirway1 (p=0.1292)
ALS2 | 226291_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8517)
ALS2 | 226291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.332)
ALS2 | 226291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.669)
ALS2 | 226291_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2268)
ALS2 | 226291_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALS2 | 226291_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4916
p=0.5421
p=0.7909
p=0.2244
p=0.7831
p=0.9299
p=0.3024
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2